

GENX: Porewater Methane

May 2026 Samples

2026-06-24

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

Run Information: Brenden D. Blakley

```
#set the run date & user name
sample_year <- "2026"
sample_month <- "May"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20260618_GHGdata_GENX_May_Run1.csv","Raw Data/20260623_GHGdata_GENX_May_Run2.csv")

# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/GENX_ShimadzuGC_QAQC.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_GENX_Porewater_CH4_202605.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)

#record any notes about the run or anything other info here:
run_notes <- "Sample Ch1-80 not collected"

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

Sample Ch1-80 not collected

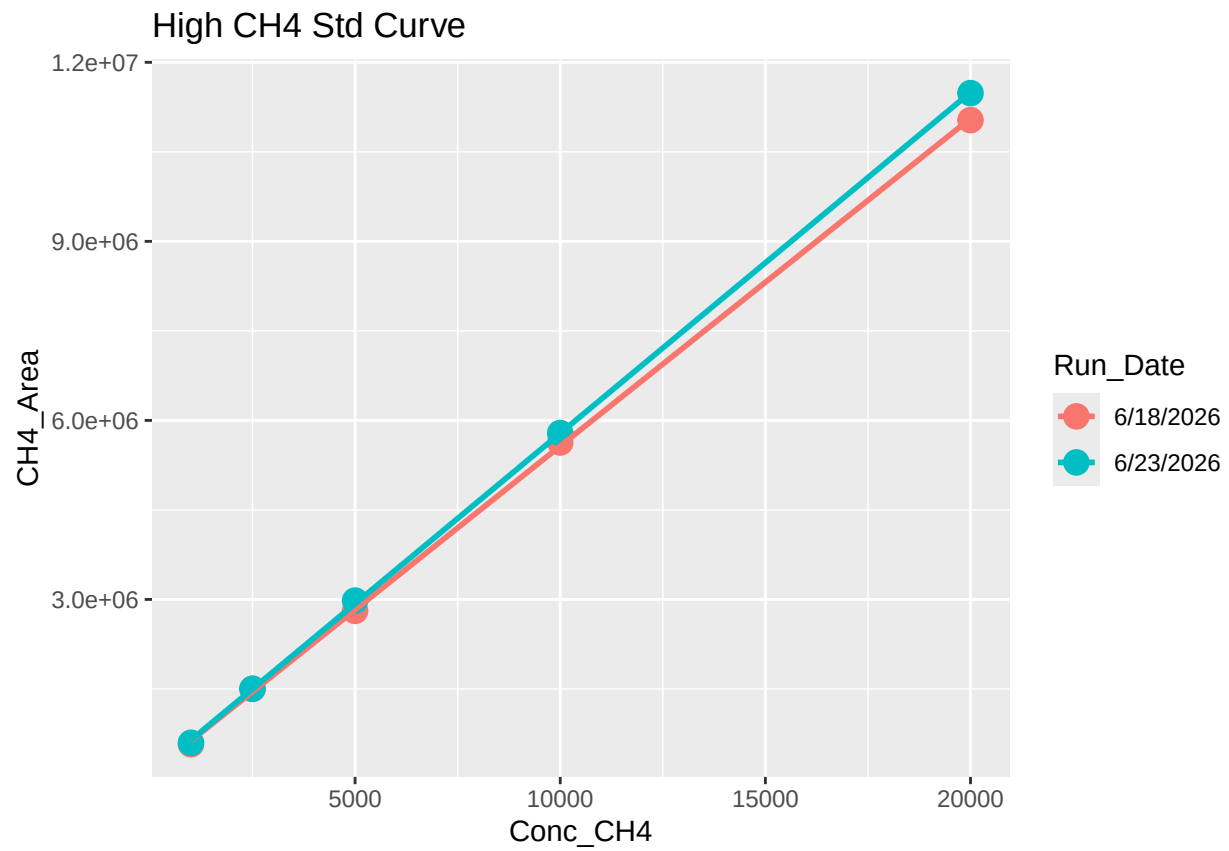
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

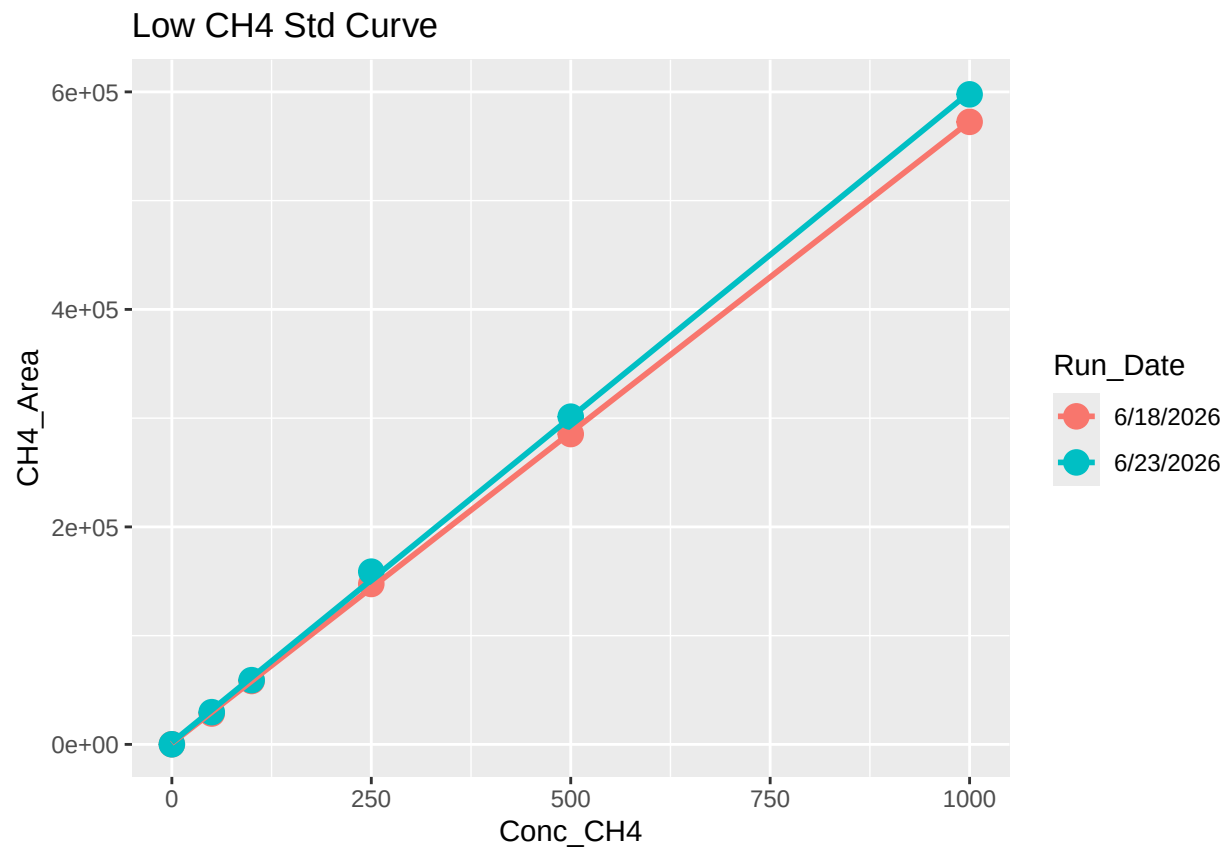
1 Read in data files

2 Assess standard curves

'geom_smooth()' using formula = 'y ~ x'

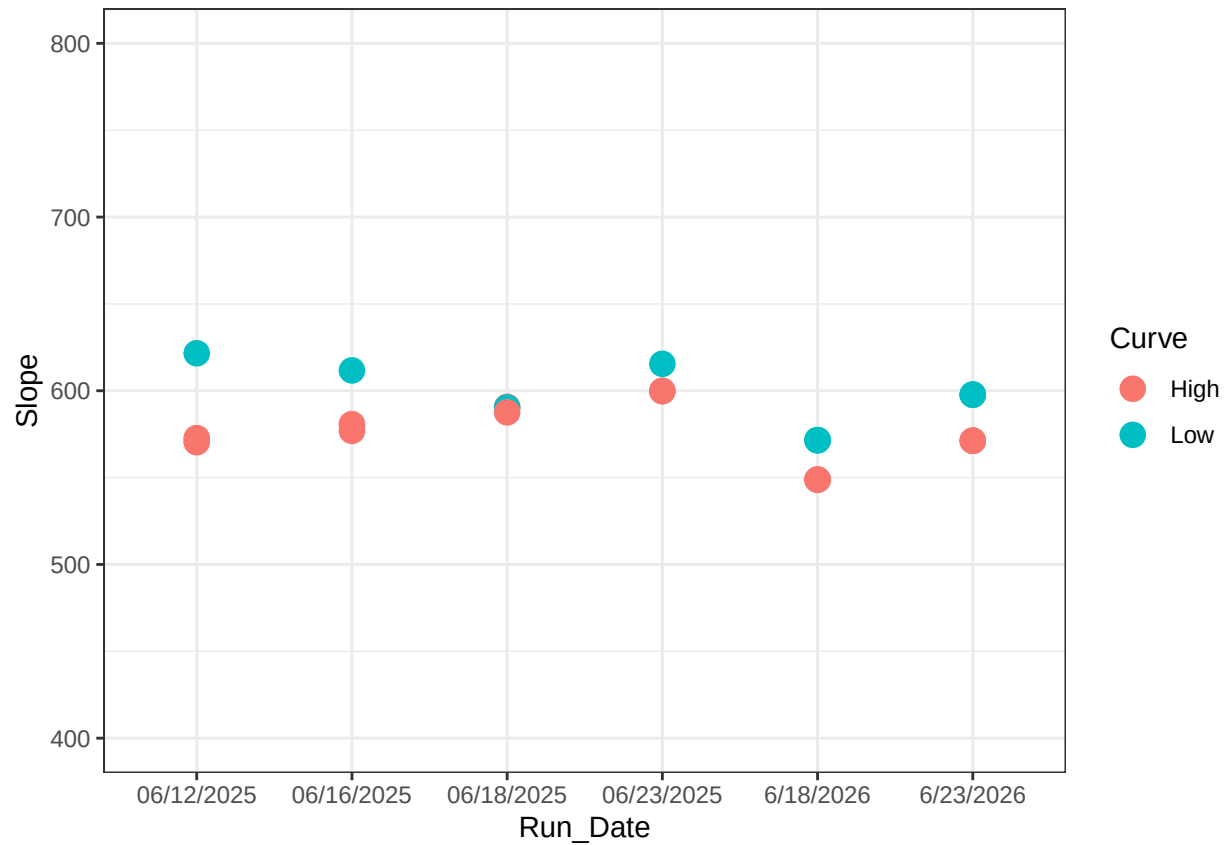


```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'
```

##	X	Curve	R2	Slope	Intercept	Run_Date
## 1	1	High	0.9986295	576.8990	173986.2174	06/16/2025
## 2	2	Low	0.9990260	611.6961	-3791.2684	06/16/2025
## 3	3	High	0.9999101	570.3925	138747.3478	06/12/2025
## 4	4	Low	0.9979098	621.5734	3224.2535	06/12/2025
## 5	5	High	0.9989882	590.7374	-109278.7391	06/18/2025
## 6	6	Low	0.9990995	590.3138	954.7682	06/18/2025



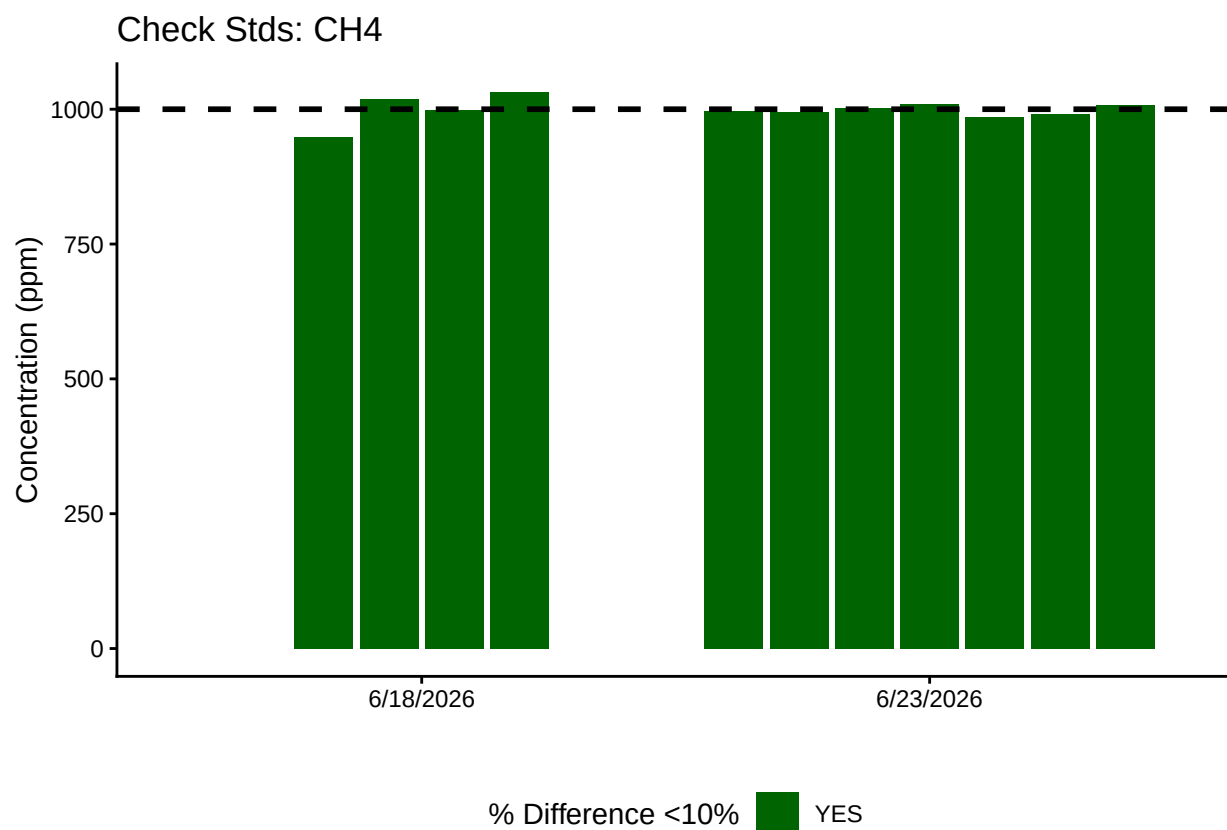
```
## [1] "High_R2 is Good-6/18/2026" "High_R2 is Good-6/23/2026"
```

```
## [1] "Low_R2 is Good-6/18/2026" "Low_R2 is Good-6/23/2026"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

Assess Blanks

>60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

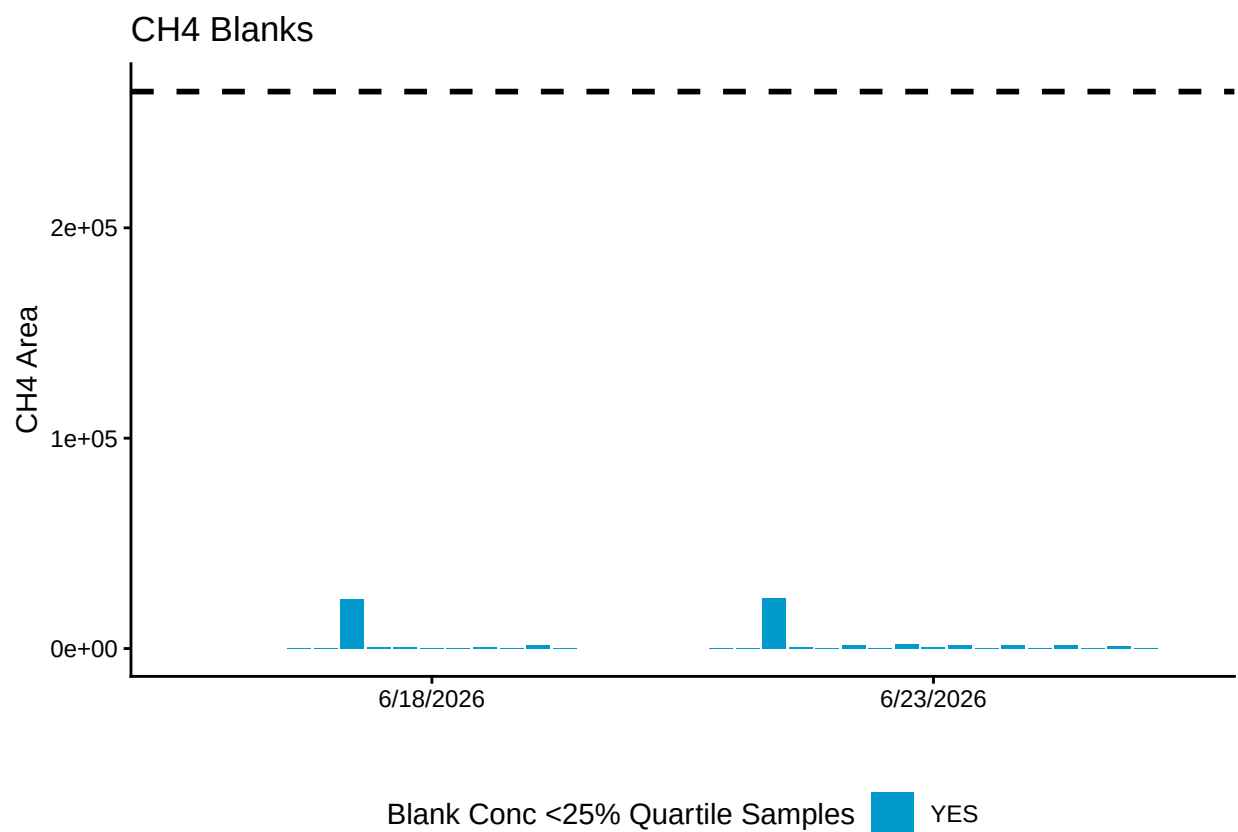
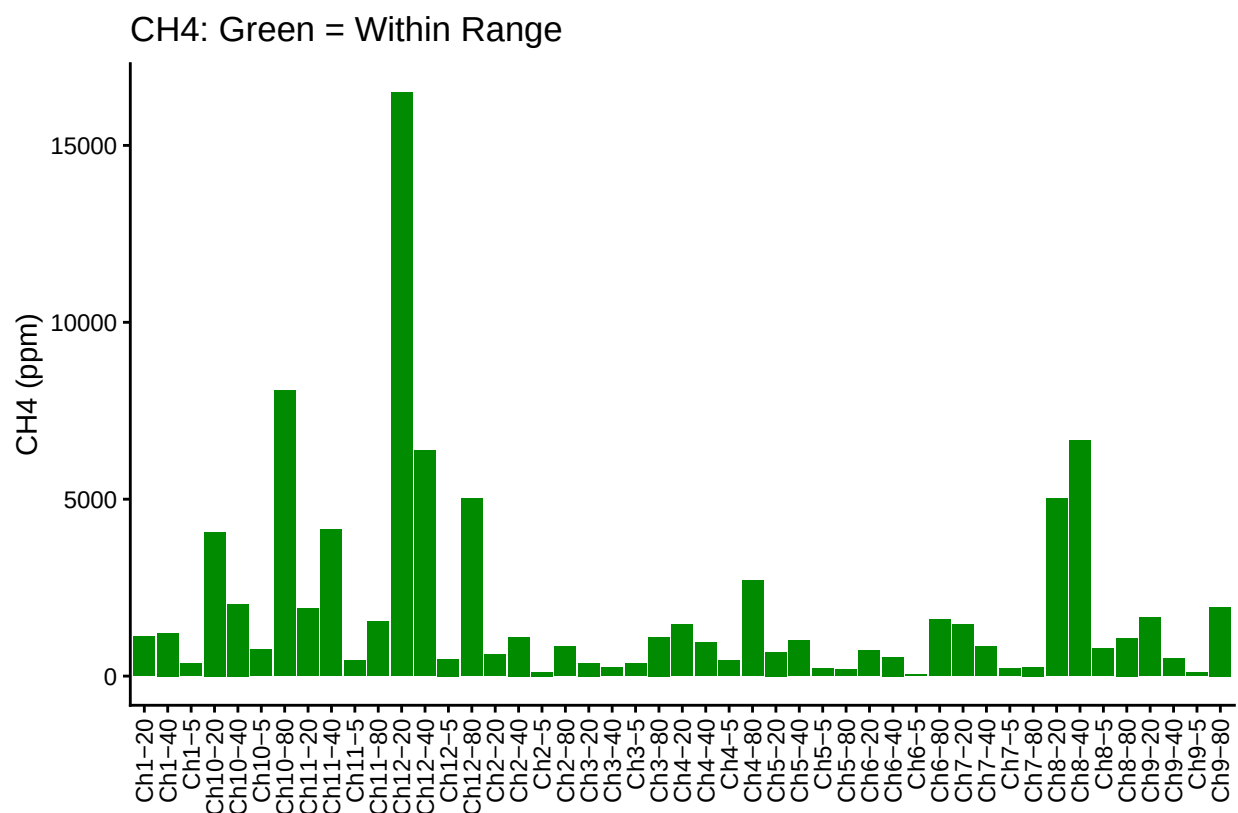
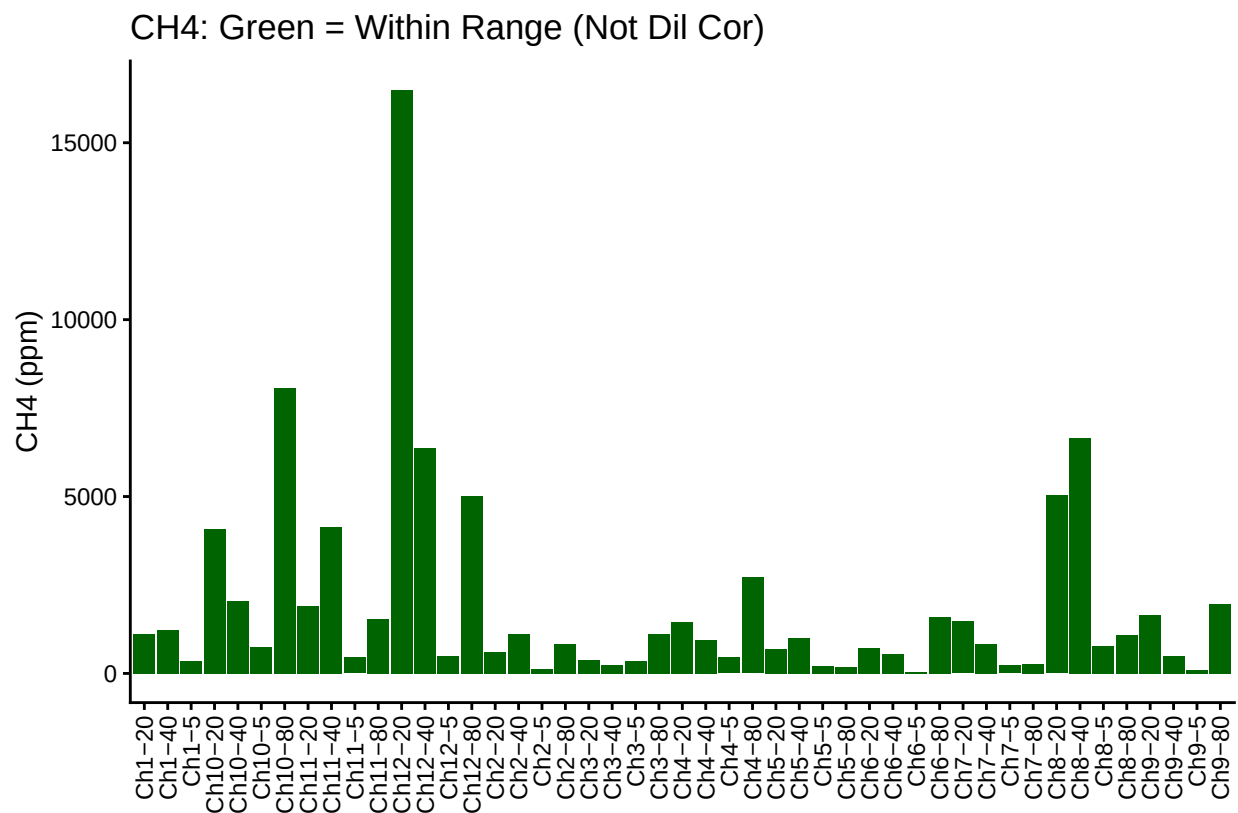


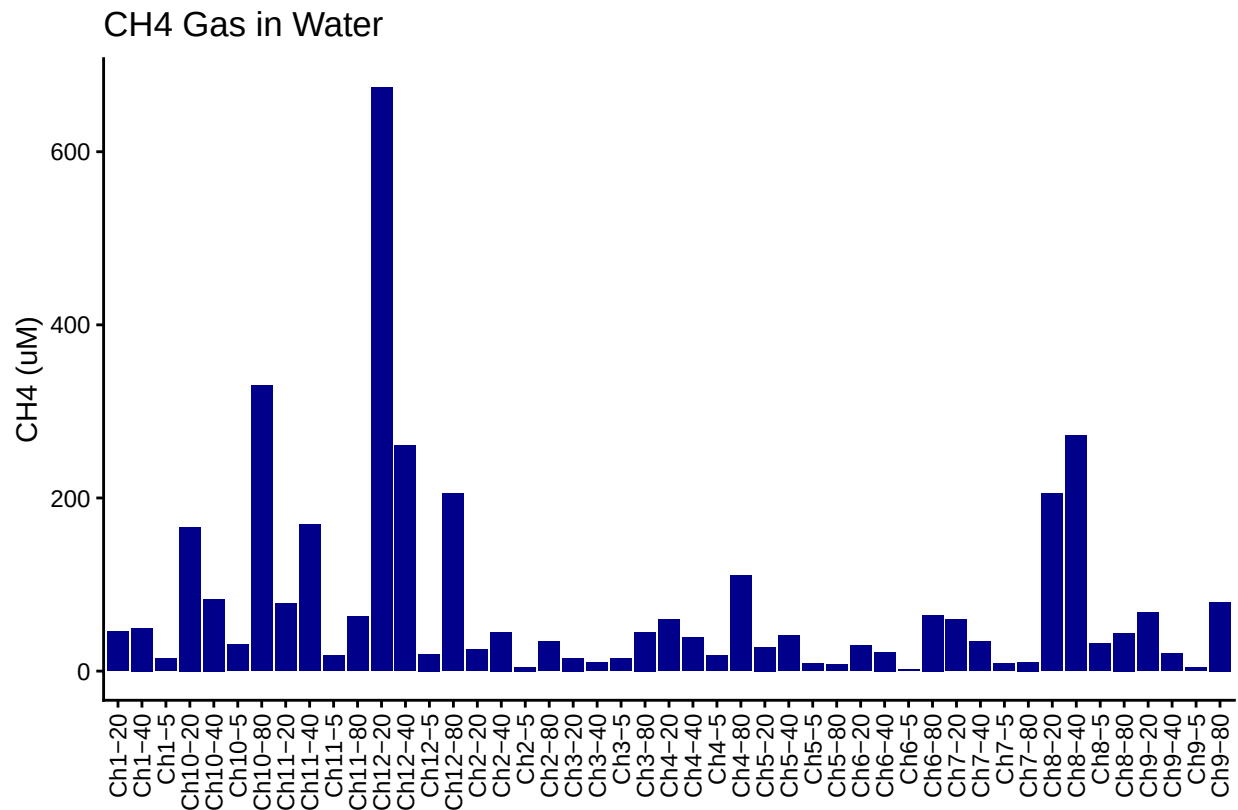
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	2366.357

5 Dilution correct samples and Check Range



6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

```
## [1] "Ch1-80"
```

7.1 Organize Data

```
## Run_Date Curve Sample_ID Type Conc_CO2 Conc_CH4 Conc_N2O CO2_Area CH4_Area
## 1 6/18/2026 High Ch1-20 Sample unknown unknown unknown NA 696817
## 2 6/18/2026 High Ch1-40 Sample unknown unknown unknown NA 751656
## 3 6/18/2026 High Ch2-40 Sample unknown unknown unknown NA 687944
## 4 6/18/2026 High Ch3-80 Sample unknown unknown unknown NA 688223
## 5 6/18/2026 High Ch7-20 Sample unknown unknown unknown NA 887157
## 6 6/18/2026 High Ch4-20 Sample unknown unknown unknown NA 877497
## N2O_Area Notes Sample_Volume R2 Slope Intercept Min Max
## 1 NA NA 12 0.9998111 548.8349 82289.68 50 20000
## 2 NA NA 12 0.9998111 548.8349 82289.68 50 20000
```

```

## 3      NA      NA      12 0.9998111 548.8349 82289.68 50 20000
## 4      NA      NA      12 0.9998111 548.8349 82289.68 50 20000
## 5      NA      NA      12 0.9998111 548.8349 82289.68 50 20000
## 6      NA      NA      12 0.9998111 548.8349 82289.68 50 20000
##  CH4_Conc_ppm      CH4_Flag CH4_Conc_ppm_dilcorr CH4_Conc_uM
## 1      1119.694 Within Range      1119.694      45.76497
## 2      1219.613 Within Range      1219.613      49.84893
## 3      1103.527 Within Range      1103.527      45.10418
## 4      1104.036 Within Range      1104.036      45.12496
## 5      1466.502 Within Range      1466.502      59.93993
## 6      1448.901 Within Range      1448.901      59.22053

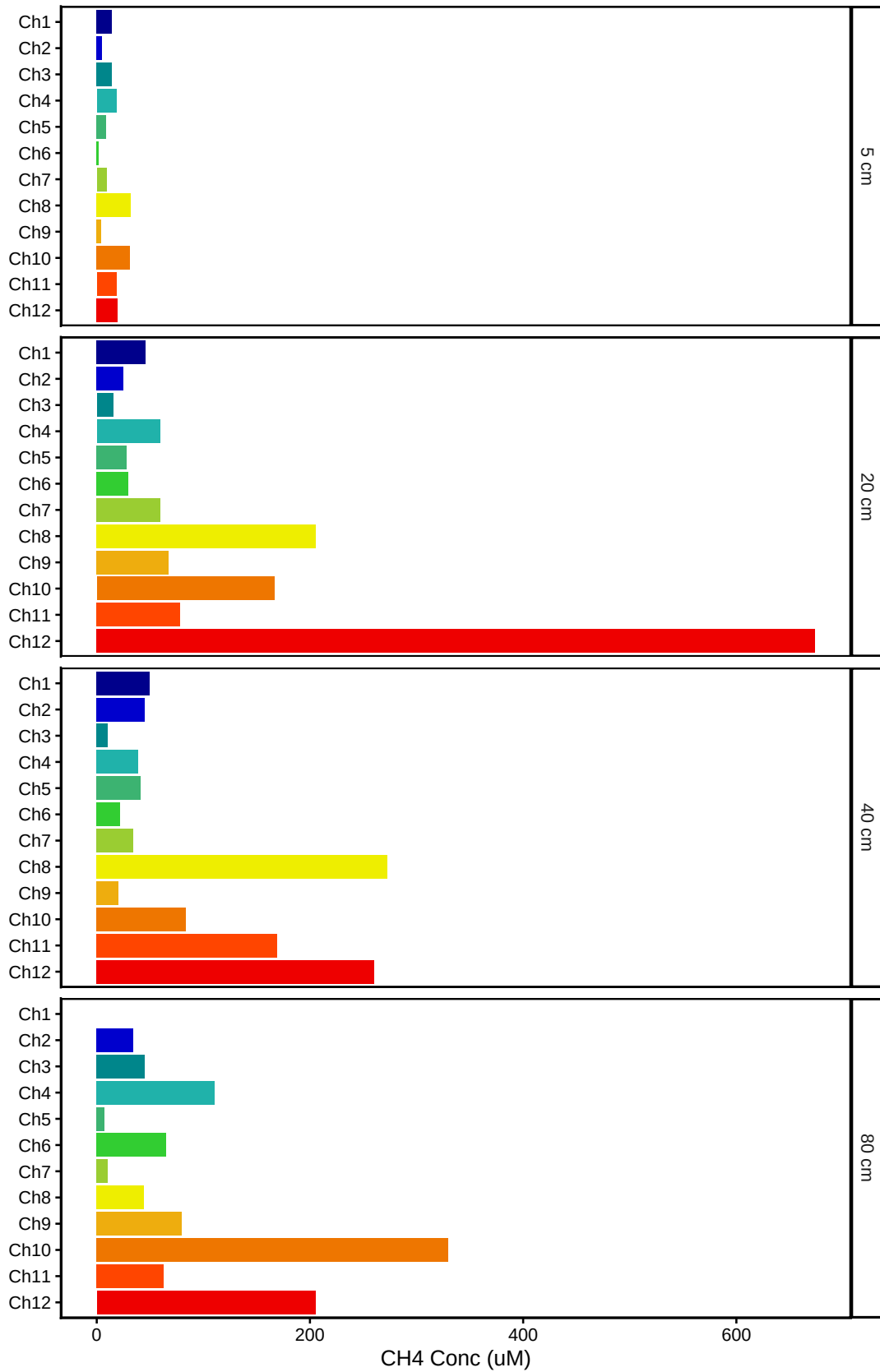
##  Site Project Year Month      Community Nickname Plot Depth_cm
## 1 GCRew      GENX 2026      May SCAM, SPPA, DISP      GENX Ch1      20
## 2 GCRew      GENX 2026      May SCAM, SPPA, DISP      GENX Ch1      40
## 3 GCRew      GENX 2026      May SCAM, SPPA, DISP      GENX Ch2      40
## 4 GCRew      GENX 2026      May SCAM, SPPA, DISP      GENX Ch3      80
## 5 GCRew      GENX 2026      May SCAM, SPPA, DISP      GENX Ch7      20
## 6 GCRew      GENX 2026      May SCAM, SPPA, DISP      GENX Ch4      20
##  CO2_Treatment Temp_Treatment CH4_Conc_ppm CH4_Conc_uM CH4_Conc_Flag
## 1      Ambient      0.00      1119.694      45.76497 Within Range
## 2      Ambient      0.00      1219.613      49.84893 Within Range
## 3      Ambient      0.00      1103.527      45.10418 Within Range
## 4      Ambient      0.75      1104.036      45.12496 Within Range
## 5      Ambient      3.00      1466.502      59.93993 Within Range
## 6      Ambient      1.50      1448.901      59.22053 Within Range
##  CH4_QAQC_Flags Analysis_rundate      Run_notes
## 1      6/18/2026 Sample Ch1-80 not collected
## 2      6/18/2026 Sample Ch1-80 not collected
## 3      6/18/2026 Sample Ch1-80 not collected
## 4      6/18/2026 Sample Ch1-80 not collected
## 5      6/18/2026 Sample Ch1-80 not collected
## 6      6/18/2026 Sample Ch1-80 not collected

```

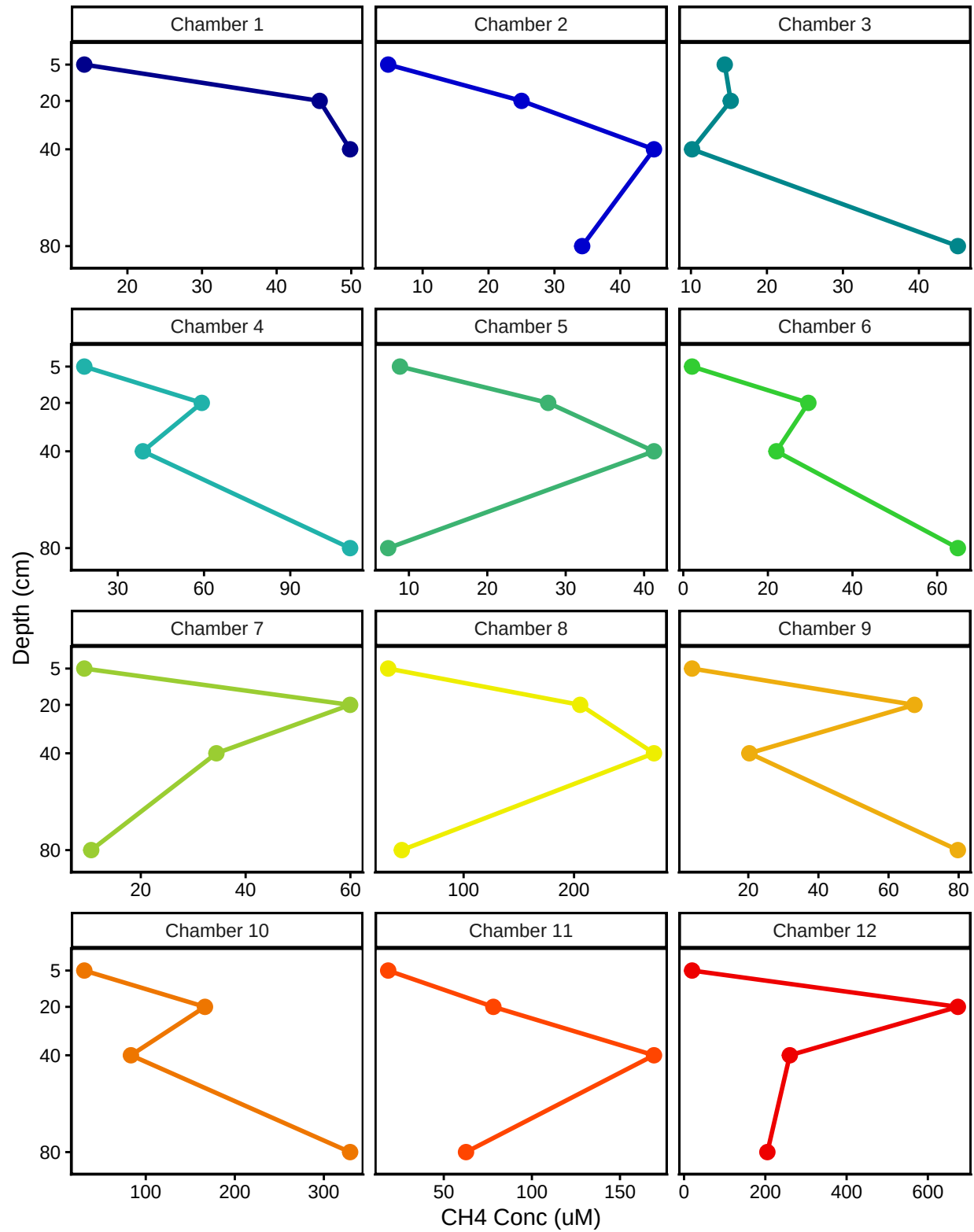
7.2 Visualize Data CH4

```
## Visualize Data
```

GENX: Porewater CH4



GENX: Porewater CH4 by Chamber



##Export Data

#end